

Part XI Dermatology - Chapter 11F

Dermatology Review: Integrated Rash Cases, Diagnostic Algorithms, Treatment Traps, and Red Flags

This review chapter synthesizes the dermatology overview, inflammatory dermatoses, infectious dermatoses, bullous and drug reactions, pigment/hair/nail disorders, and skin cancer topics into practical diagnostic pathways. The purpose is to make the dermatology section usable for shelf-style reasoning, ward triage, and future Graph-RAG extraction. The core strategy is to start with morphology, then refine by distribution, timing, symptoms, host risk, exposure, and danger signs.

Dermatology often feels like pattern recognition, but safe pattern recognition is structured. A rash should be triaged first for danger, then classified by lesion type, then linked to a small differential. A clinician should ask: Is the patient toxic? Is the rash painful or pruritic? Is it blanching or nonblanching? Are mucous membranes or eyes involved? Is there a recent drug exposure? Is the patient immunocompromised? Those questions separate benign outpatient dermatoses from emergencies such as Stevens-Johnson syndrome/toxic epidermal necrolysis, anaphylaxis, necrotizing infection, meningococcemia, vasculitis, erythroderma, and ocular zoster.

1. Universal Rash Triage: What Must Be Ruled Out First

The first question in any rash is not simply what the rash is, but whether it is dangerous today. Red flags include fever, hypotension, altered mental status, severe skin pain, rapidly progressive purpura, necrosis, bullae with skin sloughing, mucosal erosions, ocular involvement, airway symptoms, and immunosuppression. A dramatic rash can be benign if the patient is stable and the morphology fits urticaria or pityriasis rosea. A subtle rash can be lethal if it is purpuric, painful, drug-associated, or accompanied by shock.

Red flag	Why it matters	Immediate action
Fever, hypotension, confusion, or rapidly progressive rash	May reflect sepsis, toxic shock, meningococcemia, necrotizing infection, SJS/TEN, or erythroderma with fluid and temperature instability.	Full vital sign assessment, labs/cultures when indicated, urgent treatment or admission.
Mucosal erosions, conjunctivitis, severe skin pain, dusky macules, positive Nikolsky sign	Suggests SJS/TEN, pemphigus vulgaris, severe drug eruption, or extensive bullous disease.	Stop culprit drugs, urgent dermatology, ophthalmology if ocular signs, burn/ICU-level supportive care if extensive.
Palpable purpura, retiform purpura, necrosis, or digital ischemia	Suggests vasculitis, antiphospholipid syndrome, emboli, cryoglobulinemia, DIC, calciphylaxis, or severe infection.	Assess renal function/urinalysis, CBC, platelets, coags, inflammatory markers, and biopsy an active lesion if safe.
Airway symptoms with hives or angioedema	Anaphylaxis or laryngeal angioedema can be fatal even if skin findings are subtle.	IM epinephrine for anaphylaxis; airway monitoring; remove trigger.
New rash in pregnancy, neonate, transplant patient, HIV with low CD4, or chronic steroids	Host risk changes the differential and lowers threshold for disseminated infection or severe drug toxicity.	Escalate evaluation; consider disseminated HSV/VZV, fungal infection, bacterial sepsis, and medication toxicity.

2. Morphology-First Differential Diagnosis

Morphology converts a broad complaint into a manageable differential. Macules and patches are flat color changes. Papules and plaques are raised solid lesions. Vesicles and bullae are fluid-filled lesions. Pustules contain purulent or neutrophilic material. Wheals are transient edematous plaques. Scale indicates abnormal keratinization. Erosions and ulcers indicate loss of epidermis or dermis. Scratching, infection, steroid exposure, and partial treatment can obscure morphology, so ask what the first lesion looked like before it was scratched or treated.

Dominant morphology	High-yield differential	Clues that narrow the diagnosis
Comedones, papules, pustules, nodules	Acne vulgaris, rosacea, folliculitis, hidradenitis suppurativa, steroid acne, perioral dermatitis.	Comedones favor acne; central face flushing and telangiectasia favor rosacea; intertriginous abscesses and sinus tracts favor hidradenitis.
Erythematous plaques	Psoriasis, eczema/contact dermatitis, tinea corporis, seborrheic dermatitis, lichen simplex chronicus.	Extensor silvery plaques and nail pitting favor psoriasis; annular leading scale favors tinea; flexural pruritic lichenification favors eczema.
Vesicles or grouped vesicles	HSV, VZV, allergic contact dermatitis, dyshidrotic eczema, dermatitis herpetiformis, early bullous disease.	Grouped painful vesicles favor herpes; dermatomal distribution favors zoster; intensely pruritic extensor grouped papulovesicles favor dermatitis herpetiformis.

Bullae or erosions	Bullous pemphigoid, pemphigus vulgaris, SJS/TEN, burns, bullous impetigo, porphyria cutanea tarda.	Tense bullae in older adults favor bullous pemphigoid; flaccid bullae with oral erosions favor pemphigus; skin pain and drug exposure favor SJS/TEN.
Wheals or angioedema	Urticaria, anaphylaxis, ACE-inhibitor angioedema, hereditary angioedema, urticarial vasculitis.	Individual wheals lasting less than 24 hours favor urticaria; airway/GI symptoms suggest anaphylaxis; nonpruritic recurrent swelling suggests bradykinin-mediated disease.
Purpura, petechiae, necrosis	Vasculitis, thrombocytopenia, meningococemia, endocarditis, DIC, emboli, calciphylaxis.	Palpable purpura suggests small-vessel vasculitis; retiform purpura or necrosis suggests vascular occlusion.
Pigmented changing lesion	Melanoma, dysplastic nevus, seborrheic keratosis, lentigo, pigmented BCC.	ABCDE changes, ugly-duckling lesion, bleeding, rapid evolution, or firm nodularity require biopsy or referral.

3. Distribution Patterns That Matter

Distribution is often more diagnostic than a single lesion. Psoriasis classically favors extensor surfaces, scalp, nails, umbilicus, and gluteal cleft. Atopic dermatitis favors flexural surfaces in older children and adults. Seborrheic dermatitis tracks sebaceous areas such as the scalp, eyebrows, nasolabial folds, ears, central chest, axillae, and groin. Tinea corporis often expands centrifugally with peripheral scale and central clearing. Scabies favors finger webs, wrists, axillae, waistline, genital region, and household clustering. Herpes zoster respects a dermatome and usually does not cross midline. Hidradenitis suppurativa favors apocrine-bearing intertriginous areas such as axillae, groin, inframammary folds, perineum, and buttocks.

Location also changes interpretation. A hand rash may be irritant dermatitis from wet work, allergic contact dermatitis from nickel or rubber accelerators, psoriasis, tinea manuum, dyshidrotic eczema, or scabies. A facial rash may be acne, rosacea, seborrheic dermatitis, lupus, perioral dermatitis, dermatomyositis, photosensitivity, or cosmetic contact dermatitis. A foot rash may be tinea pedis, dyshidrotic eczema, contact dermatitis, psoriasis, scabies, or vascular disease.

4. Rash Algorithms

Algorithm A - Acute generalized rash with fever

Step 1: Check vitals and toxicity. Hypotension, altered mental status, severe pain, neck stiffness, or rapidly progressive purpura are emergency findings. Step 2: Ask about new medications in the past 1 to 8 weeks, especially antibiotics, anticonvulsants, allopurinol, NSAIDs, sulfonamides, and immune checkpoint inhibitors. Step 3: Look for mucosal erosions, facial edema, lymphadenopathy, eosinophilia, target lesions, bullae, desquamation, and organ injury. Step 4: decide whether the syndrome is infectious, drug-mediated, inflammatory, or vascular. Treat sepsis and anaphylaxis immediately; stop suspicious drugs early when severe cutaneous adverse reaction is possible.

Algorithm B - Itchy scaly rash

First identify whether scale is localized, annular, flexural, extensor, sebaceous, interdigital, or generalized. Annular plaques with leading scale should have KOH testing before steroids because corticosteroids can create tinea incognito. Flexural chronic itchy plaques support atopic dermatitis; sharply demarcated plaques after exposure support contact dermatitis; extensor plaques with silvery scale and nail pitting support psoriasis; scalp or nasolabial greasy scale supports seborrheic dermatitis. If the rash is nocturnally pruritic, involves finger webs or genitals, or multiple household contacts itch, treat or test for scabies.

Algorithm C - Blistering eruption

Blistering disease is triaged by blister stability, mucosal involvement, pain, drug exposure, age, and distribution. Tense bullae in an older adult with urticarial plaques suggests bullous pemphigoid. Flaccid bullae and painful oral erosions suggest pemphigus vulgaris. Skin pain, dusky macules, mucositis, fever, and recent high-risk medication suggest SJS/TEN. Grouped vesicles suggest herpes; dermatomal grouped vesicles suggest zoster. Suspected autoimmune blistering disease usually needs one lesional biopsy for routine histology and one perilesional biopsy for direct immunofluorescence.

Algorithm D - Suspicious pigmented lesion

Use ABCDE and ugly-duckling logic, but do not rely only on size. Asymmetry, border irregularity, color variegation, diameter greater than 6 mm, and evolution are classic melanoma warning signs. A new, changing, bleeding, nodular, painful, or outlier lesion warrants biopsy. When melanoma is suspected, biopsy should obtain adequate depth and architecture, usually by narrow excisional biopsy when feasible; superficial shave of a suspicious melanoma can impair staging because Breslow depth is essential.

5. Integrated Comparison Table: Common Derm Diagnoses

Disorder	Prototype presentation	Diagnostic move	Treatment anchor	Trap
Acne vulgaris	Comedones with inflammatory papules/pustules; face, chest, back.	Clinical; assess severity, scarring, pregnancy potential, medication triggers.	Topical benzoyl peroxide and retinoid; add antibiotics for inflammatory disease; isotretinoin for severe nodulocystic disease.	Antibiotic monotherapy promotes resistance; isotretinoin is highly teratogenic.
Rosacea	Central face erythema/flushing, papules/pustules, telangiectasia; no comedones.	Clinical; ask ocular symptoms and triggers.	Avoid triggers; topical metronidazole, azelaic acid, or ivermectin; doxycycline for inflammatory lesions.	Mistaken for acne; topical steroids can worsen rosacea.
Atopic dermatitis	Chronic pruritic flexural eczema; personal/family atopy.	Clinical; consider infection or allergic contact if refractory.	Emollients, trigger control, topical steroids/calcineurin inhibitors.	Frequent bathing without moisturization worsens barrier dysfunction.
Allergic contact dermatitis	Pruritic vesicles/plaques in exposure distribution; delayed type IV reaction.	Patch testing if allergen unclear or recurrent.	Avoid allergen, topical steroids, systemic steroids for severe cases.	May appear days after exposure, so history must look backward.
Psoriasis	Well-demarcated plaques with silvery scale on extensor surfaces/scalp; nail pitting.	Clinical; screen for arthritis and metabolic/cardiovascular comorbidity.	Topicals for limited disease; phototherapy/systemics/biologics for severe disease.	Systemic steroids may precipitate rebound or erythrodermic flares.
Seborrheic dermatitis	Greasy scale in scalp, eyebrows, nasolabial folds, ears, chest.	Clinical; consider HIV/Parkinson disease if severe.	Antifungal shampoos/ketoconazole; low-potency topical steroid briefly.	Chronic relapsing disorder, not a one-time infection.
Pityriasis rosea	Herald patch followed by oval trunk lesions in Christmas-tree pattern.	Clinical; consider syphilis if palms/soles/systemic risk.	Reassurance, antihistamines/topical steroids for itch.	Can mimic tinea or secondary syphilis.
Lichen planus	Pruritic purple polygonal planar papules; wrists, shins, oral/genital mucosa.	Clinical plus/minus biopsy; consider hepatitis C/drugs.	Topical steroids; systemic therapy for severe mucosal/widespread disease.	Oral erosive disease can be chronic and painful.
Urticaria	Transient intensely pruritic wheals, individual lesions usually less than 24 h.	Clinical; evaluate anaphylaxis symptoms.	Second-generation H1 antihistamines; epinephrine if anaphylaxis.	Persistent painful lesions over 24 h may be urticarial vasculitis.
Angioedema	Deep swelling of lips, eyelids, tongue, extremities, genitalia.	Airway assessment; medication review, especially ACE inhibitor.	Epinephrine if anaphylaxis; stop ACE inhibitor; targeted therapy for hereditary/bradykinin forms.	ACE-inhibitor angioedema may occur after long uneventful use.

6. Infectious Dermatology Review

Skin infection questions are solved by combining lesion morphology with immune status and exposure. Warts are HPV-related hyperkeratotic papules; genital warts are often HPV 6/11, while oncogenic cervical disease is classically linked to HPV 16/18. Molluscum contagiosum is a poxvirus infection producing small umbilicated papules and can be extensive in immunocompromised patients. Herpes zoster presents with grouped vesicles in a dermatomal pattern, often preceded by neuropathic pain; disseminated or ophthalmic involvement requires escalation. Dermatophytes infect keratinized structures such as skin, hair, and nails; KOH is the bedside test for scale. Tinea capitis and onychomycosis usually require systemic therapy because topical therapy penetrates poorly into hair shafts and nails. Tinea versicolor is *Malassezia*-associated and produces hypo- or hyperpigmented finely scaling macules. Scabies causes severe nocturnal pruritus, burrows, and clustered household symptoms; treatment requires patient and close-contact therapy plus decontamination of bedding and clothing.

Syndrome	Pattern	Diagnostic clue/test	Treatment anchor
Verruca vulgaris/plantar/plana	Hyperkeratotic papules on hands, knees, feet, face, or legs.	Clinical; plantar lesions interrupt skin lines and may be painful.	Salicylic acid or cryotherapy; many resolve spontaneously.
Condyloma acuminatum	Soft fleshy anogenital growths.	Clinical; STI assessment and cervical cancer prevention context.	Patient-applied or clinician-applied therapy; HPV vaccination prevention.
Molluscum contagiosum	Smooth dome-shaped umbilicated papules.	Clinical; extensive disease suggests immunosuppression.	Observation, curettage, cryotherapy, cantharidin in selected settings.
Herpes zoster	Painful grouped vesicles in a dermatome.	Clinical; PCR if atypical; urgent eye evaluation for V1/ocular signs.	Acyclovir/valacyclovir/famciclovir early or severe; analgesia; vaccination prevention.
Tinea corporis/cruris/pedis	Annular or interdigital scale, erythema, pruritus.	KOH from active scaly border.	Topical terbinafine or azole; hygiene and moisture control.
Tinea capitis	Scalp scale, alopecia, black dots/kerion, lymphadenopathy.	KOH/culture; Wood lamp sometimes useful depending on species.	Oral griseofulvin or terbinafine; manage carriers in outbreaks.
Onychomycosis	Thick discolored dystrophic nails.	Confirm with KOH/culture/PAS before long systemic therapy.	Oral terbinafine commonly used when treatment needed.
Tinea versicolor	Trunk/shoulder dyspigmented fine scale.	KOH spaghetti-and-meatballs pattern.	Topical selenium sulfide/azole; oral therapy for extensive/recurrent disease.
Scabies	Nocturnal itch, burrows, finger webs/wrists/genitals; contacts itch.	Skin scraping can show mite, ova, scybala.	Permethrin 5 percent to entire body, repeat; treat close contacts; oral ivermectin when needed.

7. Severe Cutaneous Adverse Reactions and Drug Rashes

Most drug eruptions are morbilliform exanthems appearing within days to weeks of exposure and resolving after withdrawal of the culprit medication. However, dermatology review must emphasize severe cutaneous adverse reactions because delay can be fatal. SJS/TEN is suggested by skin pain, fever, dusky macules, mucosal erosions, and epidermal detachment. DRESS is suggested by fever, facial edema, lymphadenopathy, eosinophilia or atypical lymphocytes, and internal organ involvement, often with longer latency. AGEP is suggested by rapid onset of many sterile pustules with fever and neutrophilia. Fixed drug eruption recurs at the same site and often leaves hyperpigmentation. Erythroderma is not a single diagnosis but a final common pathway of drug eruption, psoriasis, eczema, CTCL, or other disease; it can cause hypothermia, fluid loss, high-output cardiac stress, and infection risk.

Reaction	Timing clue	Defining findings	Management concept
Morbilliform drug eruption	Usually days to weeks after drug start; faster with re-exposure.	Symmetric blanching macules/papules; minimal systemic illness.	Stop nonessential culprit, symptomatic therapy, monitor for red flags.
Fixed drug eruption	Hours to days; recurs at identical site.	Well-demarcated dusky plaque or blister, often lips/genitals/hands; residual hyperpigmentation.	Avoid culprit; topical steroids/symptomatic care.

SJS/TEN	Often 1-3 weeks after high-risk drug.	Fever, skin pain, dusky lesions, mucositis, epidermal sloughing.	Stop culprit immediately, urgent admission, supportive wound/fluid/eye care.
DRESS	Often 2-8 weeks after drug.	Fever, facial edema, lymphadenopathy, eosinophilia, hepatitis/nephritis/pneumonitis.	Stop culprit; systemic steroids often used for organ involvement; prolonged monitoring.
AGEP	Often rapid, sometimes within days.	Numerous sterile pustules on erythematous base, fever, neutrophilia.	Stop culprit; supportive/topical steroids; usually resolves quickly.
Erythroderma	Variable.	More than 90 percent body surface erythema/scale; thermoregulation and fluid issues.	Admit if unstable; search cause; supportive care and dermatology guidance.

8. Bullous Disease Review

Autoimmune blistering disorders are high-yield because the level of epidermal split predicts the clinical appearance. Pemphigus vulgaris is intraepidermal and acantholytic, so blisters are fragile and erosions dominate; oral mucosa is commonly involved. Bullous pemphigoid is subepidermal, so bullae are tense and often occur on an urticarial or eczematous base in older adults. Dermatitis herpetiformis is a gluten-sensitive intensely pruritic papulovesicular disease on extensor surfaces and is associated with granular IgA in dermal papillae. SJS/TEN can mimic bullous disease but is primarily a severe drug reaction with epidermal necrosis and mucosal injury.

Disease	Blister type	Mucosa	Immunology/path clue	Treatment concept
Pemphigus vulgaris	Flaccid bullae, erosions, positive Nikolsky.	Common erosions.	oral IgG against desmoglein 3 +/- 1; intercellular fishnet IgG.	Systemic steroids plus rituximab or steroid-sparing immunosuppression.
Bullous pemphigoid	Tense bullae on pruritic urticarial/eczematous plaques.	Uncommon/milder	IgG against BP180/BP230; linear basement membrane IgG/C3.	High-potency topical steroids for limited; systemic therapy for extensive.
Dermatitis herpetiformis	Excoriated grouped papulovesicles, intensely pruritic.	Usually no major mucositis.	Granular IgA in dermal papillae; celiac association.	Gluten-free diet; dapsone for rapid symptom control.
SJS/TEN	Dusky macules, atypical targets, epidermal detachment.	Prominent mucositis.	Drug-triggered keratinocyte necrosis.	Stop culprit, supportive care, ocular/burn-unit level management.

9. Skin Cancer and Precancer Review

Skin cancer review depends on recognizing risk patterns. Actinic keratoses are rough scaly sun-damaged lesions and markers of cumulative UV injury; hypertrophic, bleeding, tender, or indurated lesions require biopsy to exclude squamous cell carcinoma. Seborrheic keratoses are benign stuck-on waxy plaques, often multiple and inherited; they can mimic melanoma when pigmented or inflamed. Keratoacanthoma is a rapidly growing crateriform keratin-filled lesion that resembles SCC and is often managed with excision or biopsy because clinical distinction can be unreliable. Basal cell carcinoma classically forms pearly papules with telangiectasias or nonhealing ulcers. Squamous cell carcinoma forms scaly, crusted, keratotic, indurated papules or plaques and has higher metastatic risk in immunosuppressed patients, lip/ear/genital sites, scars, chronic wounds, and perineural invasion. Melanoma is suggested by ABCDE change, ugly-duckling pattern, nodularity, bleeding, or rapid evolution.

Lesion	Classic clue	Why it matters	Typical next step
Actinic keratosis	Rough gritty scaly lesion on sun-exposed skin.	Premalignant UV damage and SCC risk marker.	Cryotherapy for isolated lesions; field therapy for extensive damage; biopsy suspicious lesions.

Seborrheic keratosis	Stuck-on waxy/verrucous plaque, sharply demarcated.	Benign; can mimic melanoma.	Reassure if classic; remove if symptomatic/cosmetic; biopsy atypical lesions.
Basal cell carcinoma	Pearly papule, telangiectasia, rolled border, ulceration.	Locally destructive; metastasis rare.	Biopsy then excision, Mohs in high-risk sites, or other selected therapies.
Squamous cell carcinoma	Firm scaly/crusted/keratotic papule or plaque, nonhealing ulcer.	Metastatic risk higher than BCC, especially high-risk sites/hosts.	Biopsy and surgical management; assess risk features.
Melanoma	ABCDE, ugly duckling, changing pigmented lesion, nodular growth.	Depth and ulceration drive staging/prognosis.	Prompt biopsy that preserves staging information; wide excision based on pathology.
Cutaneous T-cell lymphoma	Persistent patches/plaques mimicking eczema/psoriasis, bathing-trunk distribution.	Can be missed for years; erythroderma/lymphadenopathy suggests advanced disease.	Biopsy repeated if persistent/refractory; dermatology/oncology management.

10. Bedside Tests and Biopsy Strategy

The best bedside test is chosen by the question being asked. KOH preparation answers whether a scaly eruption contains fungal elements. Mineral oil scraping or skin scraping can support scabies by demonstrating mites, eggs, or fecal pellets. Tzanck smear is historically associated with herpes but is nonspecific and largely replaced by PCR when diagnostic certainty is needed. Wood lamp can help in selected pigment and infection questions but has limitations because not all organisms fluoresce. Bacterial culture is useful for purulent drainage, recurrent abscesses, unusual cellulitis, immunocompromised hosts, and treatment failure. Viral PCR is valuable for atypical HSV/VZV, immunocompromised hosts, or severe disease.

Clinical question	Preferred test/strategy	Pearl
Is this tinea?	KOH scraping from active scaly border.	Do before topical steroids if annular scale is present.
Is this scabies?	Look for burrows; scraping/mineral oil prep if possible.	Treat close contacts simultaneously even if asymptomatic.
Is this HSV/VZV?	PCR from vesicle base if atypical/severe.	Zoster in young or disseminated disease should prompt immune-status thinking.
Is this melanoma?	Biopsy that captures full lesion and depth when feasible.	Breslow depth is central to staging.
Is this autoimmune bullous disease?	Lesional biopsy for H&E plus perilesional biopsy for DIF.	DIF site selection matters; avoid fully eroded skin for DIF.
Is this vasculitis?	Biopsy fresh purpuric lesion plus labs/urinalysis.	Skin findings may be the entry point to renal, GI, or neurologic disease.
Is this CTCL?	Multiple/repeat biopsies of persistent refractory patches/plaques.	Stop topical steroids before biopsy if possible to improve yield.

11. Treatment Traps and Safety Pearls

Trap	Why it is dangerous or misleading	Safer habit
Steroids on undiagnosed annular scale	Can mask dermatophyte infection and produce tinea incognito.	Perform KOH or treat fungus when morphology strongly suggests tinea.
Systemic steroids for psoriasis without clear indication	Withdrawal can trigger severe rebound, pustular psoriasis, or erythroderma.	Use psoriasis-directed topical/phototherapy/systemic plans; consult for severe disease.
Antibiotic monotherapy in acne	Promotes resistance and relapse.	Pair antibiotics with benzoyl peroxide and limit duration when possible.
Missing ocular zoster	V1 involvement can threaten vision.	Urgent ophthalmology when eye pain, vision change, conjunctivitis, Hutchinson sign, or V1 rash.
Treating scabies patient only	Contacts and fomites maintain transmission.	Treat close contacts and wash/heat-bag clothing/bedding.
Calling all lip/tongue swelling allergy	ACE-inhibitor/bradykinin angioedema may not respond to antihistamines/steroids like histamine disease.	Prioritize airway and medication review; stop ACE inhibitor permanently.

Ignoring mucosa in rash exam	Mucositis changes differential toward SJS/TEN, pemphigus, erythema multiforme, Behcet, infection.	Inspect mouth, eyes, genital mucosa in serious or widespread eruptions.
Reassuring a changing lesion because it is small	Small melanomas occur; evolution is critical.	Biopsy/referral for evolving or ugly-duckling lesions.

12. Integrated Case Drills

Case stem	Likely diagnosis	Key reasoning	Immediate next step
Teen with open/closed comedones and inflammatory papules on face/back.	Acne vulgaris.	Comedones anchor acne; rosacea lacks comedones.	Benzoyl peroxide plus topical retinoid; add therapy by severity.
Middle-aged patient with central facial flushing, telangiectasia, papules, no comedones.	Rosacea.	Central face plus flushing plus no comedones.	Trigger avoidance; topical therapy; assess ocular disease.
Pruritic flexural plaques in patient with asthma/allergic rhinitis.	Atopic dermatitis.	Atopy plus chronic pruritic flexural eczema.	Emollients, trigger reduction, topical anti-inflammatory therapy.
Vesicular rash on hands 2 days after poison ivy exposure.	Allergic contact dermatitis.	Delayed type IV reaction in exposure distribution.	Avoid allergen, topical steroids; systemic steroids if severe/widespread.
Extensor plaques with silvery scale and nail pitting.	Psoriasis.	Classic plaque psoriasis distribution and nail disease.	Topical steroids/vitamin D analogs if limited; screen for psoriatic arthritis.
Herald patch followed by oval trunk lesions in Christmas-tree pattern.	Pityriasis rosea.	Self-limited papulosquamous eruption.	Reassurance, pruritus control; consider syphilis if atypical.
Grouped painful vesicles in unilateral thoracic dermatome.	Herpes zoster.	Dermatomal grouped vesicles with neuropathic pain.	Antiviral therapy when early/severe/high-risk; pain control.
Nighttime itch, finger web burrows, multiple family members itching.	Scabies.	Nocturnal pruritus plus contacts plus classic distribution.	Permethrin for patient/contacts; environmental cleaning.
Annular scaly plaque with central clearing after topical steroid worsened it.	Tinea corporis/tinea incognito.	Steroids can mask or worsen dermatophyte infection.	KOH from scale; topical antifungal if localized.
Elderly patient with tense bullae on urticarial plaques, minimal mouth disease.	Bullous pemphigoid.	Subepidermal tense bullae in older adult.	Biopsy with DIF; topical/systemic therapy by severity.
Painful oral erosions and flaccid bullae.	Pemphigus vulgaris.	Fragile intraepidermal blisters plus mucosal involvement.	Urgent dermatology; biopsy/DIF; systemic immunosuppression.
Fever, skin pain, mucositis, dusky macules after allopurinol.	SJS/TEN spectrum.	High-risk drug plus mucosa plus skin pain/necrosis.	Stop drug, urgent admission/supportive care, ophthalmology.
New facial edema, fever, eosinophilia, hepatitis 6 weeks after anticonvulsant.	DRESS.	Long latency plus eosinophilia plus organ injury.	Stop culprit, monitor organs, systemic steroids if significant.
Rough gritty lesion on sun-damaged scalp.	Actinic keratosis.	Precancerous UV-damage marker.	Cryotherapy/field therapy; biopsy if indurated/tender/bleeding.
Changing asymmetric pigmented lesion with irregular border and multiple colors.	Melanoma until proven otherwise.	ABCDE evolution and color variation.	Prompt adequate biopsy/referral.

13. Final Review: Dermatology One-Page Mental Model

Step 1: Decide if it is dangerous. Look for fever, toxicity, hypotension, pain, mucosa, eyes, purpura, necrosis, bullae, rapidly progressive spread, immunosuppression, pregnancy, or high-risk medication exposure. Dangerous rashes are stabilized before they are elegantly classified.

Step 2: Name the primary lesion. Macule, patch, papule, plaque, vesicle, bulla, pustule, wheal, purpura, erosion, ulcer, scale, crust, lichenification, atrophy, or nodule. Secondary changes often obscure the initial clue, so ask where the rash started and what the first lesion looked like.

Step 3: Use distribution. Dermatomal means zoster until proven otherwise. Interdigital/nocturnal/household itching means scabies. Extensor silvery plaques means psoriasis. Flexural chronic itch means atopic dermatitis. Sebaceous greasy scale means seborrheic dermatitis. Annular leading scale means tinea. Central facial flushing without comedones means rosacea. Intertriginous

abscesses and sinus tracts mean hidradenitis.

Step 4: Ask exposure questions. New drugs, topical products, occupational wet work, plants, travel, hotel/dorm/jail/shelter exposure, sexual exposure, animal exposure, sunlight, immunosuppression, and close contacts can all be diagnostic.

Step 5: Choose targeted testing. KOH for scale, scraping for scabies, PCR for atypical herpes, culture for purulence or unusual infection, biopsy for cancer/vasculitis/bullous disease/refractory inflammatory disease. Testing should answer a specific question rather than be ordered as a reflex panel.

Step 6: Treat safely. Stop culprit drugs in severe eruptions, use epinephrine for anaphylaxis, avoid indiscriminate systemic steroids in psoriasis or undiagnosed infection, treat scabies contacts, protect vision in ocular zoster/SJS, and biopsy changing pigmented lesions promptly.

References used for clinical updating and synthesis

- American Academy of Dermatology: ABCDEs of melanoma and melanoma clinical guideline highlights.
- American Academy of Dermatology: Guidelines of care for acne vulgaris, updated 2024.
- Centers for Disease Control and Prevention: Recombinant zoster vaccine recommendations for adults 50 years and older and immunocompromised adults 19 years and older.
- Merck Manual Professional Edition: Scabies; Stevens-Johnson syndrome and toxic epidermal necrolysis; dermatologic evaluation and rash pattern references.
- Uploaded Medvale dermatology source packets derm1 and derm2, used as local coverage guides for acneiform, papulosquamous, allergic, infectious, infestation, benign, premalignant, and lesion recognition topics.